



JABchem



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Past Papers Higher Chemistry

2024 Marking Scheme

Grade Awarded	Mark Required		% candidates achieving grade
	(/150)	%	
A	101+	67%	29.7%
B	83+	55%	23.0%
C	65+	43%	21.6%
D	47+	31%	16.5%
No award	<47	<31%	9.2%

Section:	Multiple Choice	Extended Answer	Assignment
Average Mark:	15.8 /25	48.7 /95	19.4 /30

2024 Higher Chemistry Marking Scheme

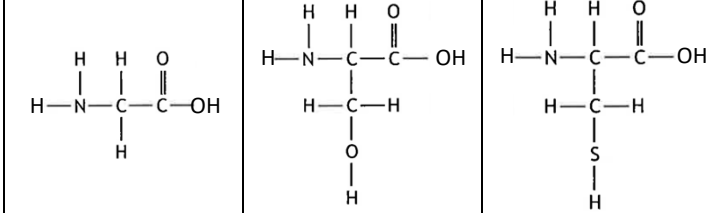
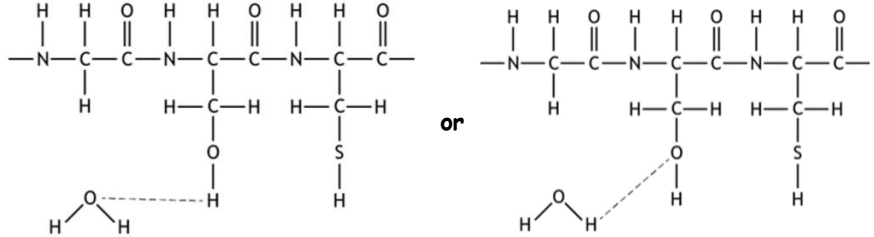
MC Qu	Answer	% Pupils Correct	Reasoning																																																
1	C	53	☒A Number of electrons in same outer electron shell does not decide the value of the first ionisation energy. ☒B Neutrons in nucleus are not involved in the removal of electrons from outer shell ☒C Chlorine has a greater nuclear charge which holds outer electrons more tightly as they are closer to nucleus. This increases the first ionisation energy of chlorine. ☒D Mass of an atom has no influence on ionisation energies																																																
2	D	48	<table><tr><th colspan="6">State at 30°C</th><th colspan="6">Covalent Radii</th></tr><tr><th colspan="6">Melting Points for Group 1 Alkali Metals:</th><th colspan="6">Covalent Radii for Group 1 Alkali Metals:</th></tr><tr><td>Li</td><td>Na</td><td>K</td><td>Rb</td><td>Cs</td><td>Fr</td><td>Li</td><td>Na</td><td>K</td><td>Rb</td><td>Cs</td><td>Fr</td></tr><tr><td>181°C</td><td>98°C</td><td>63°C</td><td>39°C</td><td>28°C</td><td>less than 28°C</td><td>130pm</td><td>160pm</td><td>200pm</td><td>215pm</td><td>238pm</td><td>greater than 238pm</td></tr></table> As you go down group 1, the melting point of the Alkali Metals decreases. - melting point of Francium must be below the 28°C melting point of Caesium - If Francium has a melting point below 28°C then Francium will be a liquid at 30°C as it will have melted at 30°C. As you go down group 1, the covalent radii of the Alkali Metals increases. - An addition electron shell (energy level) is added to each member of the alkali metals going down Group 1 - Francium has one more electron shell than caesium so has a covalent radii greater than 238pm	State at 30°C						Covalent Radii						Melting Points for Group 1 Alkali Metals:						Covalent Radii for Group 1 Alkali Metals:						Li	Na	K	Rb	Cs	Fr	Li	Na	K	Rb	Cs	Fr	181°C	98°C	63°C	39°C	28°C	less than 28°C	130pm	160pm	200pm	215pm	238pm	greater than 238pm
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3	C	62	☒A C=O bonds are polar due to electronegativity difference of 0.8 CO₂ is a non-polar molecule overall due to linear shape ☒B H-S bonds have an electronegativity difference of 0.4 and the bonds are not pure covalent as electrons are not shared completely evenly ☒C P-H bonds are non-polar due to P and H atoms having the same electronegativity Same electronegativity ☒D C-F bonds are polar due to electronegativity difference of 1.4 CF₄ is a non-polar molecule due to tetrahedral shape																																																
4	C	53	① $2\text{I}^- \rightarrow \text{I}_2 + 2\text{e}^-$ ② $\text{MnO}_4^- + 8\text{H}^+ + 5\text{e}^- \rightarrow \text{Mn}^{2+} + 4\text{H}_2\text{O}$ ① x5 $10\text{I}^- \rightarrow 5\text{I}_2 + 10\text{e}^-$ ② x2 $2\text{MnO}_4^- + 16\text{H}^+ + 10\text{e}^- \rightarrow 2\text{Mn}^{2+} + 8\text{H}_2\text{O}$ ①'+②' $2\text{MnO}_4^- + 16\text{H}^+ + 10\text{I}^- \rightarrow 2\text{Mn}^{2+} + 8\text{H}_2\text{O} + 5\text{I}_2$ 2mol 10mol 1mol 5mol																																																
5	B	52	Sulfite is in the right hand side of the electrochemical series. The equation in the electrochemical series is reversed to become an oxidation reaction. $\text{SO}_3^{2-} + \text{H}_2\text{O} \rightarrow \text{SO}_4^{2-} + 2\text{H}^+ + 2\text{e}^-$ Sulfite ions will react with a species below it on the left hand side of the electrochemical series. As sulphite ions are oxidised, it will react with an oxidising agent (the oxidising agent is reduced itself) ☒A Iodide I⁻ ions are on the right of electrochemical series ∴ no reaction ☒B Iodine I₂ (below SO₃²⁻ on the left) is oxidised into iodide ions: I₂ + 2e⁻ → 2I⁻ ☒C Sulphite SO₃²⁻ ions are oxidised so sulfite ions are acting as a reducing agent ☒D Sulfate SO₄²⁻ ions cannot react with iodide I⁻ ions as SO₄²⁻ is above I⁻ on left.																																																

6	A	56	isovaleric acid 3-methylbutanoic acid $ \begin{array}{c} \text{H} \quad \text{CH}_3 \text{H} \\ \quad \quad \\ \text{H}-\text{C}-\text{C}-\text{C}-\text{C} \begin{array}{l} \nearrow \text{O} \\ \searrow \text{OH} \end{array} \\ \quad \quad \\ \text{H} \quad \text{H} \quad \text{H} \\ (\text{CH}_3)_2\text{CHCH}_2\text{COOH} \end{array} \longrightarrow \begin{array}{c} \text{H} \quad \text{H} \quad \text{CH}_3 \\ \quad \quad \\ \text{H}-\text{C}-\text{C}-\text{C}-\text{C} \begin{array}{l} \nearrow \text{O} \\ \searrow \text{OH} \end{array} \\ \quad \quad \\ \text{H} \quad \text{H} \quad \text{H} \\ \text{CH}_3\text{CH}_2\text{CH}(\text{CH}_3)\text{COOH} \end{array} $ is an isomer of 2-methylbutanoic acid
7	C	70	☒ A $\text{C}_4\text{H}_7\text{OH}$ is unsaturated. Formula would need to be $\text{C}_4\text{H}_9\text{OH}$ to be saturated. ☒ B $\text{C}_4\text{H}_7\text{OH}$ is unsaturated. Formula would need to be $\text{C}_4\text{H}_9\text{OH}$ to be saturated. ☒ C secondary alcohols oxidise to ketones with acidified dichromate solution ☒ D tertiary alcohols do not oxidise with acidified dichromate solution
8	A	90	time = 3 minutes 20 seconds = 200 seconds $\text{rate} = \frac{1}{\text{time}} = \frac{1}{200} = 0.005 \text{ s}^{-1}$
9	B	83	☒ A Not an α -amino acid as there are two carbons between $-\text{COOH}$ and $-\text{NH}_2$ groups ☒ B one carbon group between $-\text{COOH}$ and $-\text{NH}_2$ groups $\therefore$ α -amino acid ☒ C Not an α -amino acid as there are four carbons between $-\text{COOH}$ and $-\text{NH}_2$ groups ☒ D Not an α -amino acid as there are three carbons between $-\text{COOH}$ and $-\text{NH}_2$ groups
10	A	74	☒ A Isoprene C_5 unit is on the end of C_{10} terpene ☒ B not an isoprene group as C_5 group is in the middle of C_{10} terpene ☒ C not an isoprene group as group only has four carbons. Isoprene has 5 carbons. ☒ D not an isoprene group as group only has four carbons. Isoprene has 5 carbons
11	B	61	The higher the iodine number the greater the number of $\text{C}=\text{C}$ double bonds. - the greater the number of $\text{C}=\text{C}$ double bonds the higher the degree of unsaturation and the further apart the edible oil molecules are - the further apart the edible oil molecules the lower the melting point.
12	C	59	☒ A propan-2-ol produces only one product on dehydration: propene ☒ B pentan-3-ol produces only one product on dehydration: pent-2-ene ☒ C hexan-3-ol produces two products on dehydration: hex-2-ene and hex-3-ene ☒ D heptan-4-ol produces only one product on dehydration: hept-4-ene
13	D	54	☒ A Condensation: small molecules join together with small molecule/water removed ☒ B Hydrolysis: larger molecule splits with small molecule/water added across break ☒ C Oxidation: increase in the oxygen : hydrogen ratio e.g. primary alcohol to carboxylic acid ☒ D Reduction: decrease in the oxygen : hydrogen ratio e.g. carboxylic acid to primary alcohol
14	C	72	☒ A covalent bonding is inside molecules not between molecules ☒ B alkanes are non-polar $\therefore$ no hydrogen bonding ☒ C as alkanes are non-polar, the longer the chain length the greater the strength of LDF due to the increased number of electrons in bigger molecules ☒ D alkanes are non-polar $\therefore$ no permanent dipole to permanent dipole interactions
15	A	67	☒ A catalysts do not increase energy of molecule to give more successful collisions ☒ B catalysts do lower the energy that molecules need for successful collisions ☒ C catalysts do provide an alternative reaction pathway to the products ☒ D catalysts do increase the rate of reaction
16	D	44	Ionic formula of $(\text{NH}_4)_3\text{PO}_4 = (\text{NH}_4^+)_3\text{PO}_4^{3-}$ $\therefore$ 1 mol $(\text{NH}_4)_3\text{PO}_4 = 3$ mol NH_4^+ positive ions $\therefore$ 5 mol $(\text{NH}_4)_3\text{PO}_4 = 15$ mol NH_4^+ positive ions
17	D	43	☒ A X-Y would give the activation energy of the forward reaction ☒ B Y-X would give a negative value and activation energies are always positive ☒ C Y-Z is the enthalpy change for the reverse reaction as it gives a positive value ☒ D Z-Y is the enthalpy change for the forward reaction as it gives a negative value

18	B	75	$\begin{array}{ccc} & \xrightarrow{X} & \\ \text{C(g)} + \text{O}_2\text{(g)} & & \text{C(g)} + \text{O}_2\text{(g)} \\ & \searrow \Delta H_1 = -393.5 \text{ kJ mol}^{-1} & \swarrow \Delta H_2 = -283.0 \text{ kJ} \\ & \text{CO}_2\text{(g)} & \end{array}$ $X = \Delta H_1 - \Delta H_2$ $X = -393.5 - (-283.0)$ $X = -110.5 \text{ kJ mol}^{-1}$
19	A	47	☒ A 3xN-H bonds formed in reaction $\therefore \frac{1}{3}$ of total is enthalpy for 1xN-H bond ☒ B Bonds in N_2 and H_2 must be broken before N-H bonds in NH_3 are formed ☒ C Bonds in N_2 and H_2 must be broken before N-H bonds in NH_3 are formed ☒ D Bonds in H_2 must be broken before N-H bonds in NH_3 are formed
20	D	83	As the reaction is already at equilibrium and the conditions have not changed, the rate of the forward reaction and reverse reactions will be the same.
21	A	79	$\textcircled{1}$ $\textcircled{2}$ $\textcircled{3}$ $\textcircled{1} \times -1$ $\textcircled{2}$ $\textcircled{3}$ Add $\textcircled{1} + \textcircled{2} + \textcircled{3}$ $\text{HCOOH} + \frac{1}{2}\text{O}_2 \rightarrow \text{CO}_2 + \text{H}_2\text{O}$ $\text{C} + \text{O}_2 \rightarrow \text{CO}_2$ $\text{H}_2 + \frac{1}{2}\text{O}_2 \rightarrow \text{H}_2\text{O}$ $\text{CO}_2 + \text{H}_2\text{O} \rightarrow \text{HCOOH} + \frac{1}{2}\text{O}_2$ $\text{C} + \text{O}_2 \rightarrow \text{CO}_2$ $\text{H}_2 + \frac{1}{2}\text{O}_2 \rightarrow \text{H}_2\text{O}$ $\text{C} + \text{H}_2 + \text{O}_2 \rightarrow \text{HCOOH}$ $\Delta H = b$ $\Delta H = c$ $\Delta H = d$ $\Delta H = -b$ $\Delta H = c$ $\Delta H = d$ $\Delta H = c + d - b = a$
22	B	42	☒ A X is the residue which gets trapped in the filter paper during filtration ☒ B X is the residue and Y is the filtrate ☒ C X is the residue which gets trapped in the filter paper during filtration ☒ D Y is the filtrate which goes through the filter paper during filtration
23	C	41	☒ A 50cm ³ could not be measured accurately in a beaker ☒ B 50cm ³ pipette can only measure exactly 50cm ³ ☒ C 45cm ³ of solution could be measured accurately in a 50cm ³ burette ☒ D 45cm ³ could not be measured as accurately in a measuring cylinder as in a burette
24	B	86	☒ A Repeating the titration is necessary but not for observing the end-point ☒ B A white tile helps observation of exact point where colour change takes place ☒ C Rinsing the flask between is necessary but not for observing the end-point ☒ D Disregarding the rough titre is necessary but not for observing the end-point
25	B	48	☒ A Bubbling CO_2 through calcium hydroxide (lime water) would remove the CO_2 but CO_2 would not be able to be collected. ☒ B Collection over water would isolate CO_2 from the reactants & other products ☒ C CO_2 is already a gas so evaporation is not suitable ☒ D Distillation separates liquids with different boiling points

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Long Qu	Answer	Reasoning																		
1a(i)	Answer to include:	Energy required to remove 1 mole of electrons from 1 mole of atoms in the gaseous state.																		
1a(ii)	$P(g) \rightarrow P^+(g) + e^-$	1 st Ionisation energy is the removal of one mole of electrons and. Turns gaseous atoms into gaseous one positive ions. Must be in the gaseous state.																		
1b	Answer to include:	<table><tr><td>1 mark</td><td>Stronger</td><td> intermolecular van der Waals' </td><td>forces between phosphorus compared with nitrogen</td></tr><tr><td>1 mark</td><td colspan="3">London dispersion forces (LDFs) are the intermolecular forces present.</td></tr><tr><td>1 mark</td><td colspan="3">There are more electrons in phosphorus (P₄) compared to nitrogen (N₂)</td></tr></table>	1 mark	Stronger	intermolecular van der Waals'	forces between phosphorus compared with nitrogen	1 mark	London dispersion forces (LDFs) are the intermolecular forces present.			1 mark	There are more electrons in phosphorus (P ₄) compared to nitrogen (N ₂)								
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1c	230	<table><tr><th colspan="3">Bond Breaking Steps</th><th colspan="3">Bond Forming Steps</th></tr><tr><th>No. of bonds</th><th>Energy</th><th>Total</th><th>No. of bonds</th><th>Energy</th><th>Total</th></tr><tr><td>6x P-P</td><td>6x 201</td><td>1206 kJ</td><td>2x P≡P</td><td>2x 488</td><td>976 kJ</td></tr></table> Enthalpy change = ΣBond breaking steps - Σbond forming steps Enthalpy change = 1206kJ - 976kJ Enthalpy change = 230kJ mol⁻¹	Bond Breaking Steps			Bond Forming Steps			No. of bonds	Energy	Total	No. of bonds	Energy	Total	6x P-P	6x 201	1206 kJ	2x P≡P	2x 488	976 kJ
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1d(i)	77.1%	no. of mol = $\frac{\text{mass}}{\text{gfm}} = \frac{750}{310.3} = 2.42 \text{ mol}$ $2\text{Ca}_3(\text{PO}_4)_2 + 6\text{SiO}_2 + 10\text{C} \longrightarrow 6\text{CaSiO}_3 + 10\text{CO} + \text{P}_4$ 2mol 2.42mol 1mol 1.21mol mass = no. of mol x gfm = 1.21 mol x 124.0 = 149.9g (theoretical) % Yield = $\frac{\text{actual}}{\text{theoretical}} \times 100 = \frac{115.5}{149.9} \times 100 = 77.1\%$																		
1d(ii)	reducing agent	A reducing agent is a species which reduces another species in a reaction while being oxidised itself																		
1d(iii)	147 litres	no. of mol = $\frac{\text{mass}}{\text{gfm}} = \frac{280}{28.0} = 10.0 \text{ mol}$ Volume = 1450 litres Molar Volume = $\frac{\text{Volume}}{\text{no. of mol}} = \frac{1470}{10.0} = 147 \text{ litres mol}^{-1}$																		
2a(i)	activation energy	The activation energy is the minimum energy required in a reaction to form an activated complex and then form products.																		
2a(ii)	-52.4	gfm NaN₃ = 65.0g no. of mol = $\frac{\text{mass}}{\text{gfm}} = \frac{80}{65.0} = 1.23 \text{ mol}$ 1 mol NaN₃ releases -42.6kJ 1.23 mol NaN₃ releases -42.6kJ x $\frac{1.23}{1}$ = -52.4 kJ																		
2a(iii)	one negative or 1-	As Na carries a one positive charge as a Na ⁺ ion, the negative ion must carry a one negative charge making the azide group ion N ₃ ⁻																		
2b	balanced equation showing:	$10\text{Na} + 2\text{KNO}_3 \longrightarrow \text{K}_2\text{O} + 5\text{Na}_2\text{O} + \text{N}_2$																		
2c	Answer to include:	<table><tr><td>1 mark</td><td>Silicon dioxide is a covalent network</td></tr><tr><td>1 mark</td><td>(Strong) covalent bonds in network are broken</td></tr></table>	1 mark	Silicon dioxide is a covalent network	1 mark	(Strong) covalent bonds in network are broken														
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2d	47.3	$\text{gfm NaN}_3 = 65.0\text{g}$ $\text{no. of mol} = \frac{\text{mass}}{\text{gfm}} = \frac{80}{65.0} = 1.23 \text{ mol}$ sodium azide $\longrightarrow$ nitrogen $\begin{array}{ccc} 1\text{mol} & & 1.6\text{mol} \\ 1.23\text{mol} & & 1.6\text{mol} \times 1.23/1 \\ & & = 1.97\text{mol} \end{array}$ $\text{Volume} = \text{no. of mol} \times \text{Molar Volume} = 1.97 \times 24 = 47.3 \text{ litres}$		
2e	Answer to include:	1 mark	powders have large/greater surface area	
		1 mark	more successful collisions	
3	Open Question Answer to Include:	3 mark answer	2 mark answer	1 mark answer
	Candidate Marks:	Demonstrates a <u>good</u> understanding of the chemistry involved. A good comprehension of the chemistry has provided in a logically correct, including a statement of the principles involved and the application of these to respond to the problem.	Demonstrates a <u>reasonable</u> understanding of the chemistry involved, making some statement(s) which are relevant to the situation, showing that the problem is understood.	Demonstrates a <u>limited</u> understanding of the chemistry involved. The candidate has made some statement(s) which are relevant to the situation, showing that at least a little of the chemistry within the problem is understood.
		3 marks	2 marks	1 mark
		11%	27%	34%
		0 marks	29%	
4a(i)	Water	The joining together of amino acids to form proteins is a condensation reaction. In a condensation reaction, a small molecule is removed and in the formation of proteins, water is removed as the amino acids join together.		
4a(ii)	One answer from:	Answer	Oxidation	Condensation
		Reasoning	Increase in oxygen : hydrogen ratio by removal of hydrogen	Joining together with small molecule removed at join
4a(iii)	An amino acid which can only be obtained through your diet.	Essential amino acids are the amino acids the body needs to make a range of proteins but the essential amino acids have to be taken into the body via your diet and cannot be made in the body.		
4a(iv)A	Denaturing	Denaturing occurs when the shape of a protein/enzyme is irreversibly changed and will not return to its original shape.		
4a(iv)B	one answer from:			
4a(iv)C	One dotted line as shown:			
4b(i)	0.467	$R_f = \frac{\text{distance travelled by amino acid}}{\text{distance travelled by solvent front}} = \frac{5.6}{12.0} = 0.467$		
4b(ii)	One answer from:	The spots for glycine and lysine overlap/are close together The R_f values are similar/close together for glycine and lysine.		
4b(iii)	Threonine	Threonine is the most polar as it is the only amino acid shown with a polar -OH hydroxyl group on the side group. Each of the amino acids have polar -NH ₂ and -COOH groups so the polar effect of these groups cancel each other out.		
5a	One answer from:	Fats are a concentrated source of energy	Fats are essential for the transport/storage of fat-soluble vitamins in the body	

5b(i)	$\begin{array}{c} \text{H} & \text{H} & \text{H} \\ & & \\ \text{H}-\text{C}-\text{C}-\text{C}-\text{H} \\ & & \\ \text{OH} & \text{OH} & \text{OH} \end{array}$	Glycerol has the systematic name propane-1,2,3-triol			
5b(ii)	Ester link	$\begin{array}{c} \text{O} \\ \\ -\text{C}-\text{OH} \end{array} + \text{H}-\text{O}- \longrightarrow \begin{array}{c} \text{O} \\ \\ -\text{C}-\text{O}- \end{array}$	carboxyl group	hydroxyl group	ester link
5c	Answer to include:	1 mark	Fats are more saturated (with more regular shape)		
		1 mark	They can pack together more closely/more compact structure		
		1 mark	Stronger LDF/forces of attraction between the molecules		
5d(i)	Rancid	Oxygen can react with C=C double bonds in fats/oils and the products have an unpleasant smell known as rancidity.			
5d(ii)	One answer from:	Bromine solution will quickly/rapidly decolourise with molecule B.	Bromine solution will decolourise with molecule B but not molecule A	Bromine solution will not decolourise with molecule A but will with molecule B.	
5d(iii)A	Atom or Molecule with unpaired electron	Free radicals are formed when light helps splits a covalent bond giving each end an unpaired electron. Free radicals are very reactive species.			
5d(iii)B	Initiation	Step	Reactants (before Arrow)	Products (after Arrow)	
		Initiation	No free radicals on Reactant Side	Free radicals on Product Side	
		Propagation	Free Radicals found on both sides of arrow		
		Termination	Free radicals on Reactant Side	No free radicals on Product Side	
5d(iv)A	React with free radicals forming stable molecules.	Free radical Scavenger are reactive chemicals that quickly react with any free radical species in the body. The less the number of free radicals, the less the free radical chain reactions.			
5d(iv)B	Answer to include:	1 mark	Vitamin E is less polar (than vitamin C)		
		1 mark	An explanation that links solubility of vitamin E to the polarity of fats and oils		
5e(i)	$\text{C}_{13}\text{H}_{27}\text{COONa}$ or $\text{C}_{13}\text{H}_{27}\text{COO}^-\text{Na}^+$	Soaps are formed by the neutralisation of carboxylic/fatty acids with alkalis: $\text{C}_{13}\text{H}_{27}\text{COOH} + \text{NaOH} \longrightarrow \text{C}_{13}\text{H}_{27}\text{COO}^-\text{Na}^+ + \text{H}_2\text{O}$			
5e(ii)	To prevent scum formed	In areas of hard water detergent and Ca^{2+} ions form a precipitate known as scum			
5e(iii)A	2-methylbuta-1,3-diene	$\begin{array}{c} \text{H} \\ \\ \text{H}-\text{C}-\text{H} \\ \\ \text{C} \\ // \quad \backslash \\ \text{H}-\text{C} \quad \text{C}-\text{H} \\ \quad \\ \text{H} \quad \text{H} \end{array} \text{ or } \begin{array}{c} \text{H}_2\text{C}^{\textcircled{1}}=\text{C}^{\textcircled{2}}-\text{CH}^{\textcircled{3}}=\text{CH}_2^{\textcircled{4}} \\ \\ \text{H}_3\text{C} \end{array}$			
5e(iii)B	3	Formula of farnesene = $\text{C}_{15}\text{H}_{24}$ Each isoprene unit contains 5 carbons $\therefore$ 3 isoprene units in $\text{C}_{15}\text{H}_{24}$			

6a(i)	methyl octanoate	$\text{CH}_3(\text{CH}_2)_6\text{COOCH}_3$ has the structure: 8 carbon carboxylic acid side (2nd part of ester name) 1 carbon alcohol side (1st part of ester name) = -octanoate = methyl-																															
6a(ii)A	answer to include:	<table><tr><td>1 mark</td><td>Diagram shows a labelled safe method of heating (no flames) and labelled reaction mixture.</td></tr><tr><td>1 mark</td><td>Diagram shows a labelled method of preventing the vapours escaping e.g. wet paper towel small test tube of cold water condenser at neck of test tube</td></tr></table>	1 mark	Diagram shows a labelled safe method of heating (no flames) and labelled reaction mixture.	1 mark	Diagram shows a labelled method of preventing the vapours escaping e.g. wet paper towel small test tube of cold water condenser at neck of test tube	cotton wool wet paper towel elastic band test tube beaker hot water alcohol + carboxylic acid + catalyst																										
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1 mark	Diagram shows a labelled method of preventing the vapours escaping e.g. wet paper towel small test tube of cold water condenser at neck of test tube																																
6a(ii)B	diagram showing:	The peak drawn with dotted line shows a higher activation energy. potential energy reaction pathway The line drawn by the candidate must start and finish at the same potential energies and positions on the reaction pathway.																															
6b(i)	One set from:	<table><tr><td>1 mark</td><td colspan="3">Correctly identifies molecule has two parts with different polarities</td></tr><tr><td rowspan="2">1 mark</td><td>Correctly identifies</td><td> polar part hydroxyl groups </td><td> dissolves in water dissolves in polar liquids are hydrophilic </td></tr><tr><td>Correctly identifies</td><td> non-polar part hydrocarbon chain </td><td> dissolves in biodiesel dissolves in non-polar liquids are hydrophobic </td></tr><tr><td colspan="4" style="text-align: center;">and</td></tr><tr><td colspan="4" style="text-align: center;">or</td></tr><tr><td>1 mark</td><td colspan="3">Correctly identifies molecule has two parts as hydrophilic and hydrophobic.</td></tr><tr><td rowspan="2">1 mark</td><td>Correctly identifies</td><td> hydrophilic part </td><td> dissolves in water dissolves in polar liquids </td></tr><tr><td>Correctly identifies</td><td> hydrophobic part </td><td> dissolves in biodiesel dissolves in non-polar liquids </td></tr></table>		1 mark	Correctly identifies molecule has two parts with different polarities			1 mark	Correctly identifies	polar part hydroxyl groups	dissolves in water dissolves in polar liquids are hydrophilic	Correctly identifies	non-polar part hydrocarbon chain	dissolves in biodiesel dissolves in non-polar liquids are hydrophobic	and				or				1 mark	Correctly identifies molecule has two parts as hydrophilic and hydrophobic.			1 mark	Correctly identifies	hydrophilic part	dissolves in water dissolves in polar liquids	Correctly identifies	hydrophobic part	dissolves in biodiesel dissolves in non-polar liquids
1 mark	Correctly identifies molecule has two parts with different polarities																																
1 mark	Correctly identifies	polar part hydroxyl groups	dissolves in water dissolves in polar liquids are hydrophilic																														
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	Correctly identifies	hydrophobic part	dissolves in biodiesel dissolves in non-polar liquids																														
6b(ii)	Reacting edible oils with glycerol	glycerol edible oil emulsifier + H₂O																															

6c(i)	One answer from:	<table><tr><td>size of wick</td><td>use same draught shield</td><td>material/type of container</td></tr><tr><td>stirring water in both</td><td>distance between burner/wick/flame & container</td><td>indication of same insulation in both experiments</td></tr></table>	size of wick	use same draught shield	material/type of container	stirring water in both	distance between burner/wick/flame & container	indication of same insulation in both experiments		
size of wick	use same draught shield	material/type of container								
stirring water in both	distance between burner/wick/flame & container	indication of same insulation in both experiments								
6c(ii)	One answer from:	<table><tr><td>use a draught shield</td><td>move flame closer to container</td><td>fit a lid on the container</td><td>insulate container</td></tr></table>	use a draught shield	move flame closer to container	fit a lid on the container	insulate container				
use a draught shield	move flame closer to container	fit a lid on the container	insulate container							
6c(iii)	41.8 or -41.8	Change in Temperature = $38^{\circ}\text{C} - 17^{\circ}\text{C} = 21^{\circ}\text{C}$ Heat Energy = Specific Heat Capacity $\times$ Mass $\times$ Change In Temperature $E_h = c \times m \times \Delta T$ $E_h = 4.18 \text{ kJ kg}^{-1} \text{ }^{\circ}\text{C}^{-1} \times 0.2\text{kg} \times 21^{\circ}\text{C}$ $E_h = 17.556 \text{ kJ}$ 0.420g biodiesel $\longleftrightarrow$ 17.556 kJ 1g biodiesel $\longleftrightarrow$ $17.556 \text{ kJ} \times \frac{1}{0.420}$ $= 41.8 \text{ kJ g}^{-1}$								
6d	£12.75	20% of 75 litres of E20 = $\frac{20}{100} \times 75 \text{ litres} = 15 \text{ litres of biodiesel.}$ 1 litre biodiesel $\longleftrightarrow$ £0.85 15 litres biodiesel $\longleftrightarrow$ $£0.85 \times \frac{15}{1} = £12.75$								
7a	19.4	Salinity = Chloride Ion concentration $\times$ 0.0018066 Chloride Ion concentration = $\frac{\text{Salinity}}{0.0018066}$ Chloride Ion concentration = $\frac{35}{0.0018066}$ Chloride Ion concentration = 19373 mg l^{-1} $= 19.4 \text{ g l}^{-1}$								
7b(i)A	One from:	Solution of $\left\{ \begin{array}{l} \text{accurately} \\ \text{precisely} \\ \text{exactly} \end{array} \right\}$ known concentration								
7b(i)B	Answer to include:	<table><tr><td>1 mark</td><td colspan="3">Rinse with seawater/sample/solution</td></tr><tr><td>1 mark</td><td>Fill above the mark and allow liquid to drop down to the mark (and read from the bottom of the meniscus)</td><td>or</td><td>Fill to mark, read from the bottom of the meniscus</td></tr></table>	1 mark	Rinse with seawater/sample/solution			1 mark	Fill above the mark and allow liquid to drop down to the mark (and read from the bottom of the meniscus)	or	Fill to mark, read from the bottom of the meniscus
1 mark	Rinse with seawater/sample/solution									
1 mark	Fill above the mark and allow liquid to drop down to the mark (and read from the bottom of the meniscus)	or	Fill to mark, read from the bottom of the meniscus							
7b(ii)	To identify the end point	An indicator is used to show the end point of a reaction. The colour change of the indicator can literally be caused by one drop of reactant so care is needed when approaching the end point. (Some reactions are self-indicating and need no indicator e.g. acidified potassium permanganate)								
7b(iii)A	One answer from:	<table><tr><td>only concordant values are used in average</td><td>sample 1 is not concordant.</td></tr></table>	only concordant values are used in average	sample 1 is not concordant.						
only concordant values are used in average	sample 1 is not concordant.									
7b(iii)B	0.5775	no. of mol = volume $\times$ concentration = $0.01155\text{litres} \times 0.5\text{mol l}^{-1} = 0.005775\text{mol}$ Ag^+ ion react with Cl^- ions <table><tr><td>1mol</td><td></td><td>1mol</td></tr><tr><td>0.005775mol</td><td></td><td>0.005775mol</td></tr></table> 10.0cm³ of seawater contains 0.005775mol Cl^- ions 1000cm³ of seawater contains $0.005775\text{mol } \text{Cl}^- \text{ ions} \times \frac{1000}{10.0}$ $= 0.5775 \text{ mol l}^{-1}$	1mol		1mol	0.005775mol		0.005775mol		
1mol		1mol								
0.005775mol		0.005775mol								
7b(iii)C	17.47	10cm³ of different seawater contains 0.00492mol Cl^- ions 1000cm³ of different seawater contains $0.00492\text{mol } \text{Cl}^- \text{ ions} \times \frac{1000}{10}$ $= 0.492\text{mol } \text{Cl}^- \text{ ions}$ mass. = no. of mol $\times$ gfm = $0.492 \times 35.5 = 17.466\text{g}$								

7c	Answer to include:	1 mark	Chloromethane has permanent dipole to permanent dipole interactions whereas water has hydrogen bonding		
		1 mark	Hydrogen bonding is stronger than permanent dipole to permanent dipole interactions		
8	Open Question:	3 mark answer		2 mark answer	1 mark answer
		Demonstrates a <u>good</u> understanding of the chemistry involved. A good comprehension of the chemistry has provided in a logically correct, including a statement of the principles involved and the application of these to respond to the problem.		Demonstrates a <u>reasonable</u> understanding of the chemistry involved, making some statement(s) which are relevant to the situation, showing that the problem is understood.	Demonstrates a <u>limited</u> understanding of the chemistry involved. The candidate has made some statement(s) which are relevant to the situation, showing that at least a little of the chemistry within the problem is understood.
		Candidate Marks:			
		3 marks	2 marks	1 mark	0 marks
		10%	21%	26%	42%
9a(i)	2843.55	mass of $\text{TiO}_2 = 1600\text{kg} = 1600000\text{g}$ gfm $\text{TiO}_2 = 79.9\text{g}$ no. of mol = $\frac{\text{mass}}{\text{gfm}} = \frac{1600000}{79.9} = 20025\text{ mol}$ $\begin{array}{ccccccc} \text{TiO}_2 & + & 2\text{Cl}_2 & + & \text{C} & \longrightarrow & \text{TiCl}_4 + \text{CO}_2 \\ 1\text{mol} & & 2\text{mol} & & & & \\ 20025\text{mol} & & 40050\text{mol} & & & & \end{array}$ gfm $\text{Cl}_2 = 71.0\text{g}$ mass = no. of mol $\times$ gfm = $40050 \times 71.0 = 2843554\text{g}$ = 2843.554kg			
9a(ii)	Covalent molecular	If Titanium Chloride were to follow the convention that ionic compounds are formed when a metal and non-metal combine then titanium would be a solid at room temperature. Because titanium chloride is a liquid at room temperature then we can conclude that titanium chloride is covalent molecular.			
9b(i)	20.1%	atom economy = $\frac{\text{mass of useful products}}{\text{total mass of reactants}} \times 100 = \frac{(1 \times 47.9)}{(1 \times 189.9) + (2 \times 24.3)} \times 100 = 20.1\%$			
9b(ii)	$\text{Mg} \longrightarrow \text{Mg}^{2+} + 2\text{e}^-$	This oxidation reaction is on p13 of data booklet but reversed as a reduction reaction. Oxidation reactions have electrons after the arrow Magnesium forms a $2+$ ion as magnesium atoms have electron arrangement of 2,8,2 and two electrons are lost to become the Mg^{2+} ion with the stable electron arrangement of 2,8.			
9b(iii)	10854	mass of Mg available = $750\text{kg} = 750000\text{g}$ gfm Mg = 24.3 no. of mol = $\frac{\text{mass}}{\text{gfm}} = \frac{750000}{24.3} = 30864\text{ mol}$ mass of $\text{TiCl}_4 = 1900\text{kg} = 1900000\text{g}$ gfm Mg = 189.9 no. of mol = $\frac{\text{mass}}{\text{gfm}} = \frac{1900000}{189.9} = 10005\text{ mol}$ $\begin{array}{ccccccc} \text{TiCl}_4 & + & 2\text{Mg} & \longrightarrow & \text{Ti} & + & \text{MgCl}_2 \\ 1\text{mol} & & 2\text{mol} & & & & \\ 10005\text{mol} & & 20010\text{mol (reacted)} & & & & \end{array}$ no. of mol Mg unreacted = $30864\text{mol} - 20010\text{mol} = 10854\text{mol Mg}$			
9b(iv)	Argon is unreactive	Any oxygen present in this reaction is going to react vigorously with magnesium so replacing air with argon removes the possibility of magnesium reacting with oxygen			
9c	One answer from:		can be carried out at lower temperature	requires less energy	has a higher atom economy

10a	6.0 - 4.5	From p17 of data booklet: <table><tr><th>Chemical Group</th><th>Chemical Shift Values</th></tr><tr><td>$\text{-}\overset{ }{\text{C}}=\text{CH}_2$, $\text{-}\overset{ }{\text{C}}=\overset{ }{\text{CH}}$</td><td><table><tr><td>6.0-4.5</td></tr></table></td></tr></table>		Chemical Group	Chemical Shift Values	$\text{-}\overset{ }{\text{C}}=\text{CH}_2$, $\text{-}\overset{ }{\text{C}}=\overset{ }{\text{CH}}$	<table><tr><td>6.0-4.5</td></tr></table>	6.0-4.5										
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6.0-4.5																		
10b	propanone (or acetone)	Possible structures of $\text{C}_3\text{H}_6\text{O}$ are: <table><tr><th>Propanal</th><th>Propanone</th></tr><tr><td></td><td></td></tr><tr><td>There should be 2 vertical lines for propanal corresponding to<table><tr><td>$\text{CH}_3\text{-R}$</td><td>$\text{-CH}_2\text{-CHO}$</td></tr><tr><td>1.5-0.9</td><td>2.7-2.0</td></tr></table></td><td>There should be 1 vertical line for propanone corresponding to<table><tr><td>$\text{CH}_3\text{-C=O}$</td><td>O=C-CH_3</td></tr><tr><td>2.7-2.0</td><td>2.7-2.0</td></tr></table> There is only one vertical line as the two side are identical. </td></tr></table>		Propanal	Propanone			There should be 2 vertical lines for propanal corresponding to <table><tr><td>$\text{CH}_3\text{-R}$</td><td>$\text{-CH}_2\text{-CHO}$</td></tr><tr><td>1.5-0.9</td><td>2.7-2.0</td></tr></table>	$\text{CH}_3\text{-R}$	$\text{-CH}_2\text{-CHO}$	1.5-0.9	2.7-2.0	There should be 1 vertical line for propanone corresponding to <table><tr><td>$\text{CH}_3\text{-C=O}$</td><td>O=C-CH_3</td></tr><tr><td>2.7-2.0</td><td>2.7-2.0</td></tr></table> There is only one vertical line as the two side are identical.	$\text{CH}_3\text{-C=O}$	O=C-CH_3	2.7-2.0	2.7-2.0	
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$\text{CH}_3\text{-C=O}$	O=C-CH_3																	
2.7-2.0	2.7-2.0																	
10c		<table><tr><th>Structure</th><td colspan="2"></td></tr><tr><td>Chemical Group</td><td>$\text{CH}_3\text{-R}$</td><td>$\text{R-CH}_2\text{-X}$</td></tr><tr><td>Chemical Shift</td><td>1.5-0.9</td><td>4.2-2.2</td></tr><tr><td>No of hydrogens</td><td>3</td><td>2</td></tr><tr><td>Relative Height</td><td>3</td><td>2</td></tr></table>		Structure			Chemical Group	$\text{CH}_3\text{-R}$	$\text{R-CH}_2\text{-X}$	Chemical Shift	1.5-0.9	4.2-2.2	No of hydrogens	3	2	Relative Height	3	2
Structure																		
Chemical Group	$\text{CH}_3\text{-R}$	$\text{R-CH}_2\text{-X}$																
Chemical Shift	1.5-0.9	4.2-2.2																
No of hydrogens	3	2																
Relative Height	3	2																

2024 Higher Chemistry Section 1 Multiple Choice Analysis

Item	Key	A	B	C	D	Missing	Invalid
1	C	44	1	53	2	0	0
2	D	8	7	36	48	0	0
3	C	17	7	62	13	0	0
4	C	6	31	53	11	0	0
5	B	13	52	16	19	0	0
6	A	56	31	7	6	0	0
7	C	20	4	70	6	0	0
8	A	90	3	5	2	0	0
9	B	8	83	4	5	0	0
10	A	74	7	5	14	0	0
11	B	1	61	36	3	0	0
12	C	16	17	59	8	0	0
13	D	10	14	22	54	0	0
14	C	14	4	72	10	0	0
15	A	67	18	12	3	0	0
16	D	22	19	14	44	0	0
17	D	6	5	46	43	0	0
18	B	8	75	13	4	0	0
19	A	47	20	19	13	2	0
20	D	8	2	6	83	0	0
21	A	79	9	8	4	0	0
22	B	16	42	24	18	1	0
23	C	1	34	41	25	0	0
24	B	6	86	4	4	0	0
25	B	23	48	12	16	0	0

2024 Higher Chemistry Section 2 Marks Data Analysis

2024 Item	Max Mark	Average Mark	% Average Mark	Discrimination	% Not Attempted
Q1ai	1	0.5	45.5%	42	1.0%
Q1aai	1	0.8	76.5%	25	0.8%
Q1b	3	1.8	59.0%	57	1.7%
Q1c	2	1.4	68.7%	50	1.8%
Q1di	2	1.4	68.0%	57	1.2%
Q1dii	1	0.3	26.7%	34	7.8%
Q1diii	2	1.5	75.8%	54	5.1%
Q2ai	1	0.9	91.3%	33	1.9%
Q2aai	1	0.5	45.6%	47	7.6%
Q2aiii	1	0.3	26.9%	31	2.6%
Q2b	1	0.7	67.3%	30	1.0%
Q2c	2	0.8	40.5%	51	3.1%
Q2d	2	1.3	65.4%	56	3.9%
Q2e	2	1.1	57.4%	43	1.6%
Q3	3	1.2	39.9%	66	10.2%
Q4ai	1	0.7	68.8%	56	5.6%
Q4aai	1	0.6	58.7%	35	2.8%
Q4aiii	1	0.8	82.7%	45	1.7%
Q4aivA	1	0.8	79.2%	42	8.0%
Q4aivB	1	0.5	54.2%	63	1.7%
Q4aivC	1	0.3	25.7%	37	4.4%
Q4bi	1	0.6	63.3%	37	0.5%
Q4bii	1	0.2	22.0%	20	6.1%
Q4biii	1	0.7	66.8%	30	0.6%
Q5a	1	0.2	21.2%	39	5.8%
Q5bi	1	0.7	74.8%	51	4.0%
Q5bii	1	0.4	39.1%	45	1.8%
Q5c	3	1.4	47.4%	59	3.6%
Q5di	1	0.7	68.4%	44	4.1%
Q5dii	1	0.3	34.7%	46	8.1%
Q5diiiA	1	0.5	48.6%	47	4.3%
Q5diiiB	1	0.8	84.0%	46	2.9%
Q5divA	1	0.4	38.6%	45	6.1%
Q5divB	2	0.3	17.1%	45	8.2%
Q5ei	1	0.2	18.6%	44	13.5%
Q5eii	1	0.7	66.8%	47	4.8%
Q5eiiiA	1	0.2	24.9%	44	9.7%
Q5eiiiB	1	0.8	84.2%	30	1.0%
Q6ai	1	0.4	38.7%	53	5.1%
Q6aiiA	2	0.8	41.7%	50	8.7%
Q6aiiB	1	0.2	19.6%	28	0.6%
Q6bi	2	0.9	46.6%	58	7.5%
Q6bii	1	0.3	35.0%	52	13.9%
Q6ci	1	0.2	23.1%	23	0.5%
Q6cii	1	0.8	75.1%	25	1.1%
Q6ciii	3	1.7	57.8%	62	2.8%
Q6d	1	0.7	69.2%	31	2.1%
Q7a	2	1.3	62.9%	49	4.4%
Q7biA	1	0.6	59.8%	47	4.6%
Q7biB	2	0.7	32.9%	45	5.3%
Q7bii	1	0.5	46.0%	43	2.0%
Q7biiiA	1	0.6	56.1%	42	2.2%
Q7biiiB	3	1.5	50.9%	64	6.2%
Q7biiiC	2	1.1	54.3%	52	6.5%
Q7c	2	0.9	46.6%	56	4.1%
Q8	3	1	33.1%	71	11.7%
Q9ai	2	1.4	68.5%	52	5.7%
Q9aai	1	0.3	27.5%	28	2.4%
Q9bi	2	1.5	75.3%	55	2.5%
Q9bii	1	0.6	64.6%	39	7.8%
Q9biii	3	1.3	43.1%	58	8.3%
Q9biv	1	0.3	30.9%	30	8.5%
Q9c	1	0.3	35.0%	37	6.6%
Q10a	1	0.8	79.6%	31	6.9%
Q10b	1	0.2	20.8%	25	5.9%
Q10c	1	0.4	44.0%	42	9.1%
Paper	95	48.7	51.2%		

2024 Higher Chemistry Section 2 Frequency of Marks Analysis

2024 Item	0	1	2	3
Q1ai	54	46		
Q1aii	23	77		
Q1b	24	14	22	39
Q1c	16	31	53	
Q1di	22	20	58	
Q1dii	73	27		
Q1diii	14	21	65	
Q2ai	9	91		
Q2aii	54	46		
Q2aiii	73	27		
Q2b	33	67		
Q2c	48	23	29	
Q2d	24	21	55	
Q2e	21	43	36	
Q3	29	34	27	11
Q4ai	31	69		
Q4aii	41	59		
Q4aiii	17	83		
Q4aivA	21	79		
Q4aivB	46	54		
Q4aivC	74	26		
Q4bi	37	63		
Q4bii	78	22		
Q4biii	33	67		
Q5a	79	21		
Q5bi	25	75		
Q5bii	61	39		
Q5c	24	26	32	17
Q5di	32	68		
Q5dii	65	35		
Q5diiiA	51	49		
Q5diiiB	16	84		
Q5divA	61	39		
Q5divB	74	18	8	
Q5ei	81	19		
Q5eii	33	67		
Q5eiiiA	75	25		
Q5eiiiB	16	84		
Q6ai	61	39		
Q6aiiA	39	39	22	
Q6aiiB	80	20		
Q6bi	37	33	30	
Q6bii	65	35		
Q6ci	77	23		
Q6cii	25	75		
Q6ciii	26	21	7	46
Q6d	31	69		
Q7a	16	41	42	
Q7biA	40	60		
Q7biB	49	35	15	
Q7bii	54	46		
Q7biiiA	44	56		
Q7biiiB	34	18	11	38
Q7biiiC	29	33	38	
Q7c	35	38	28	
Q8	42	26	21	10
Q9ai	21	22	58	
Q9aii	73	27		
Q9bi	15	20	66	
Q9bii	35	65		
Q9biii	36	20	23	21
Q9biv	69	31		
Q9c	65	35		
Q10a	20	80		
Q10b	79	21		
Q10c	56	44		

2024 Higher Chemistry Section 3 Assignment Marks Analysis

Item	Max Mark	Average Mark	% Average Mark	Discrimination
1	1	1	97.8%	14
2	3	1.7	58.0%	44
3a	1	0.4	35.5%	27
3b	1	1	95.1%	12
3c	1	0.6	57.8%	33
3d	1	0.6	60.7%	28
3e	1	1	95.6%	20
3f	1	0.5	53.6%	34
4a	1	1	98.1%	19
4b	1	0.9	88.5%	23
4c	1	0.9	85.3%	27
4d	1	0.7	65.1%	28
5	1	0.3	25.7%	33
6	1	0.6	59.2%	28
7	3	0.9	30.1%	41
8	1	1	97.4%	17
Paper	20	12.8	64.0%	

Item	Max Mark	0	1	2	3
1	1	2	98		
2	3	4	37	41	19
3a	1	65	35		
3b	1	5	95		
3c	1	42	58		
3d	1	39	61		
3e	1	4	96		
3f	1	46	54		
4a	1	2	98		
4b	1	11	89		
4c	1	15	85		
4d	1	35	65		
5	1	74	26		
6	1	41	59		
7	3	38	38	18	5
8	1	3	97		
Paper	20				

Item	Max Mark	Item Description
1	1	An aim that describes clearly the purpose of the investigation
2	3	Underlying chemistry
3a	1	A brief summary of the approach used to collect experimental data.
3b	1	Sufficient raw data from the candidate's experiment.
3c	1	Data presented in correctly produced table(s) and correct units shown for calculated values
3d	1	Values calculated correctly using a chemical relationship
3e	1	Data relevant to the experiment obtained from an internet/literature source
3f	1	A citation and reference for a source of the internet/literature data
4a	1	An appropriate format from the options of scatter graph, line graph or bar graph
4b	1	The axis/axes of the graph has/have suitable scale(s).
4c	1	The axes of the graph have suitable labels and units
4d	1	Accurately plotted data points and, where appropriate, a line of best fit.
5	1	A valid comparison of the experimental data with data from the internet/literature source
6	1	A valid conclusion that relates to the aim and is supported by all the data in the report.
7	3	A valid conclusion that relates to the aim and is supported by all the data in the report.
8	1	A valid conclusion that relates to the aim and is supported by all the data in the report.